# Matter as Memory: Quantum Measurement and Gravitational Microstates

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**Abstract**

Thermodynamic approaches to gravity, from Jacobson to Verlinde, relate spacetime dynamics to horizon entropy but do not specify the microscopic carriers of the entropy increments. We take as primitive that quantum measurement necessarily involves a physical interaction with energy–momentum transfer between system and detector. Within this setting, we model such interactions as localized Aichelburg–Sexl (AS) pp-wave perturbations of spacetime geometry. Each measurement event then induces a discrete shift in null horizon coordinates, contributing additively to horizon area, and coarse-graining over the resulting ensemble reproduces the Bekenstein–Hawking area law. A bilocal realization from our previous work using entangled photon pairs yields correlated AS shifts on opposite Rindler horizons, providing an operational realization of entanglement as spatial connectivity in the spirit of ER=EPR. Within this construction, the gravitational boundary term on the horizon is identified with the accumulated energy transfer associated with measurement events, linking matter dynamics and horizon entropy at the level of individual microstates. The horizon acts as a channel through which information passes, but it is the matter of the detectors, via accumulated quanta of action, that stores it.

## 1 Introduction

Two landmark results establish that gravity and Unruh thermodynamics are deeply intertwined. Jacobson showed that the Einstein equations can be derived from thermodynamic identities applied to local Rindler horizons, with horizon entropy playing the role of the fundamental quantity from which spacetime dynamics emerge [1]. Verlinde extended this perspective, arguing that gravitational forces arise from entropy gradients on holographic screens, recasting gravity as an entropic rather than a fundamental interaction [2]. Both frameworks, however, operate at the level of bulk thermodynamics and leave open a prior question: what physical process underlies each individual entropy increment?

We argue that the answer is quantum measurement. Each act of information acquisition by a physical observer constitutes a discrete microstate event, and the collective backreaction of such events, together with the causal correlations they induce, gives rise to gravitational effects. The argument proceeds in three steps. First, any observation is mediated by a physical interaction: information transfer necessarily involves an exchange of energy and momentum between the observed system and the detector, a discrete impulse. Second, through the equivalence principle, the impulse is locally indistinguishable from thrust in a gravitational field: no local measurement can distinguish the recoil associated with a quantum measurement event from acceleration in a gravitational potential.

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Third, when such impulses are modelled geometrically, via Aichelburg–Sexl (AS) pp-wave solutions, as we explore, they leave a geometric imprint on the horizon as energy and information pass through into the detector's matter degrees of freedom, and coarse-graining over the resulting ensemble recovers standard horizon thermodynamics.

Modeling the spacetime response to such localized energy–momentum transfer events using Aichelburg–Sexl (AS) pp-wave geometries follows the gravitational shockwave framework of Dray, 't Hooft [3], and others. Their construction uses AS solutions sourced by infalling classical particles to define horizon scattering amplitudes; the key difference here is that we use a quantum entangled pair emitted from a common ancestor, and the information acquisition framing connects the geometric shift directly to a classical bit acquired by an observer.

What distinguishes the present construction from all of these approaches is the combination of three features that do not appear together elsewhere in the literature: a non-thermal bilocal source that selects individual microstates from the Unruh ensemble rather than assuming thermality as a primitive; the identification of each microstate with a specific quantum measurement event carrying a calculable energy-momentum transfer; and an operational realization of entanglement-induced spatial separation at the single-microstate level with a concrete geometric imprint. Each of these features appears in isolation in various forms across the literature, but their combination, and in particular the grounding of horizon geometry in individual measurement events rather than bulk thermodynamic averages, appears to be new.

## 1.1 Conceptual Roadmap

Our approach stands in sharp contrast to the Diósi–Penrose objective collapse programme [4, 5], which postulates that gravity induces an objective, observer-independent collapse of the quantum state. In that framework, gravitational self-interaction is taken to define a physical timescale for wavefunction reduction, leading to a dynamical modification of standard quantum mechanics. We treat quantum measurement as a physical interaction mediated by energy–momentum transfer, and interpret any apparent reduction of the state as an effective description arising from such processes. The contrast between the two approaches is illustrated in Figure 1. The three solid arrows represent the perspectival grounding relations described above; the single dashed arrow represents the Diósi–Penrose causal postulate. It is the only relation in the diagram claimed to be objective and observer-independent, and it is the only one we do not adopt.

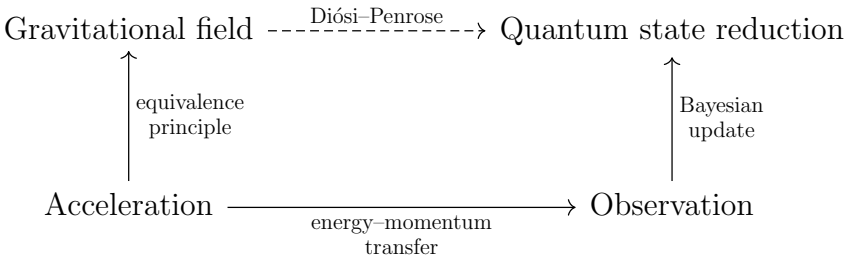


Figure 1: Solid arrows denote "substrate" relations: $A \to B$ means every instance of $B$ is physically realized by an instance of $A$, for a given observer. The dashed arrow denotes the Diósi–Penrose postulate, where the relation is claimed to hold objectively rather than observer-relatively.

To make this concrete, we build on the microstate construction of [6], which realises individual terms of the Unruh thermal ensemble via bilocal driving sources. The present

paper applies and extends that construction. Entangled pairs emitted from a localized past source propagate into left and right Rindler wedges; polarization measurement events are associated with discrete energy–momentum transfer and yield correlated classical outcomes. Each such microstate carries a calculable gravitational imprint, modelled by an AS pp-wave solution, and coarse-graining over the ensemble reproduces standard horizon thermodynamics. Our construction also provides an operational realization of entanglement-induced spatial separation at the level of individual microstates: entanglement is mapped into geometric correlations via the associated gravitational backreaction, and each measurement event contributes a discrete, calculable increment to horizon entropy through its associated energy–momentum transfer and gravitational backreaction.

# 2 Motivating Heuristics

Landauer's principle [7] is conventionally formulated for information erasure. Here we use the same energetic bound in the time-reversed operational sense: the physical registration of a measurement outcome in a detector is treated as the conjugate process to erasure, with an identical energetic gain. For an accelerated observer, the relevant temperature is the Unruh temperature [8]. Substituting one into the other eliminates temperature entirely, yielding a relation between information acquisition, energy transfer, and acceleration (a version of the black hole thermodynamics first law; equation (3) upcoming). This creates a heuristic dictionary between the thermal description natural to a Rindler observer and the non-thermal global Minkowski description, with the apparent thermality of the vacuum emerging as a consequence of the observer's acceleration rather than as a fundamental property of the state. In this way we will derive a sequence of standard results in black hole thermodynamics as a consequence of Newton's laws.

## 2.1 The Unruh-Landauer Relation

Landauer's principle suggests that the acquisition of $\Delta H$ nats of information[1] by a physical observer requires a minimum energy transfer

$$\Delta E \geq k_B T \, \Delta H. \tag{1}$$

The bound has the same form in both cases; we apply it to acquisition on the grounds that registering a measurement outcome is physically equivalent to information being written into the detector's state, which is the reverse process Landauer's principle governs.

For an observer undergoing proper acceleration $a$, the Unruh effect associates a temperature

$$T = \frac{\hbar a}{2\pi k_B c} \tag{2}$$

with the local vacuum. Substituting into Landauer's bound eliminates $k_B T$, yielding

$$\boxed{\Delta E \geq \frac{\hbar a}{2\pi c} \, \Delta H.} \tag{3}$$

Temperature has dropped out entirely: equation (3) is a temperature-independent relation between energy, acceleration, and information, suitable for the Minkowski vacuum; acceleration sets the energy scale of information.

This Unruh–Landauer bound is saturated for a photon at one bit when the wavelength equals the horizon distance up to a factor of $2\pi/\ln 2$:

$$\lambda_{\text{sat}} = \frac{2\pi}{\ln 2} \cdot \frac{c^2}{a} \sim d_{\mathcal{H}}, \tag{4}$$

[1] We are going to use $S$ for action later so use $H$ here for entropy with units of nats to avoid carrying around a $\ln(2)$.

where $d_{\mathcal{H}} = c^2/a$ is the proper distance to the Rindler horizon. The saturating photon is the fundamental mode of the causal diamond defined by the observer and the horizon, with wavelength commensurate with the horizon distance $d_{\mathcal{H}}$, too short to be super-horizon and too long to deposit more than one bit. This identifies the horizon distance as the natural length scale of a single-bit measurement event, and the Rindler horizon as the surface at which the Landauer bound is geometrically realized.

For realistic accelerations the saturation wavelength is enormous, at Earth surface gravity $a = 9.8\ \mathrm{m\,s^{-2}}$, $\lambda_{\mathrm{sat}} \approx 8$ light years, so the single-bit bound is practically always satisfied for terrestrial observers, and is only approached near astrophysical or Planck-scale accelerations.

## 2.2 Momentum Transfer and Acceleration

In what follows, standard results emerge as consequences of the Unurh-Landauer relation and Newton's laws. The energy transfer $\Delta E$ implied by the relation is carried by a physical quantum, matter or radiation, mediating the measurement interaction a *thrust quantum.* For a relativistic carrier, the associated momentum transfer is

$$\Delta p = \frac{\Delta E}{c}. \tag{5}$$

A measurement event localized to a spatial scale $r$ is bounded by a causal diamond. The temporal extent of this diamond is $\Delta t = 2r/c$, fixed entirely by the causal geometry of the event rather than by any dynamical assumption. We identify the interaction time with this natural causal duration, so that

$$a = \frac{\Delta p}{m_d \Delta t} = \frac{\Delta E}{2 m_d r}. \tag{6}$$

This perspective is closely related in spirit to Erik Verlinde's entropic gravity [2], where forces arise from entropy gradients, but differs in that the present construction grounds each entropy increment in a discrete measurement event with an associated energy-momentum transfer, rather than in a coarse-grained thermodynamic gradient.

## 2.3 The Schwarzschild Radius

We match the acceleration inferred from the measurement-induced impulse to a Newtonian gravitational field. For this identification we treat the thrust quantum as a massive ultra-relativistic particle with rest energy $\Delta E = m_t c^2$, so that $\Delta p = \Delta E/c$ holds in the limit $v \to c$, and the Newtonian gravitational field of $m_t$ at distance $r$ is well-defined. Identifying the acceleration $a$ with the Newtonian field gives:

$$a = \frac{G m_t}{r^2}, \tag{7}$$

and substituting equation (6) with $\Delta E = m_t c^2$, we obtain

$$\frac{m_t c^2}{2 m_d r} = \frac{G m_t}{r^2}. \tag{8}$$

Solving for $r$:

$$r = \frac{2 G m_d}{c^2} \equiv r_s, \tag{9}$$

which is precisely the Schwarzschild radius of the detector mass $m_d$.

It is notable that the resulting length scale is set by the detector mass $m_d$, rather than the source mass $m_t$. This reflects the fact that the construction is inherently observer-centred: the acceleration entering the argument is the response of the detector to the

measurement-induced impulse, and the associated scale is therefore the horizon radius relevant to that observer. In this sense, the quantity $r_s$ should be interpreted not as the Schwarzschild radius of the source, but as the characteristic horizon scale associated with the detector as a physical system undergoing measurement. This is consistent with the observer-dependence of horizon structure in relativistic settings, as exemplified by the Unruh effect, where acceleration alone gives rise to a causal horizon.

## 2.4 Surface Gravity

Evaluating the acceleration at $r = r_s$ gives the surface gravity of the detector:

$$\kappa = \frac{Gm_d}{r_s^2} = \frac{c^4}{4Gm_d}, \tag{10}$$

in agreement with the standard Schwarzschild result. This is the acceleration that appears in the Unruh temperature associated with the detector's horizon, $T = \hbar\kappa/2\pi k_B c$, providing an internal consistency check on the framework.

## 2.5 Bekenstein-Hawking Entropy

Returning to the Unruh-Landauer relation (3) and substituting the surface gravity $\kappa$ for $a$, the entropy increment associated with an energy increment $\Delta E = c^2 \Delta m_d$ is

$$\Delta H \leq \frac{2\pi c}{\hbar\kappa}\Delta E = \frac{8\pi G m_d}{\hbar c^3} \cdot c^2 \Delta m_d. \tag{11}$$

Expressing this in terms of the change in Schwarzschild area $\Delta A = 32\pi G^2 m_d \Delta m_d / c^4$:

$$\Delta H \leq \frac{\Delta A}{4\ell_P^2}, \qquad \ell_P^2 = \frac{\hbar G}{c^3}, \tag{12}$$

which is the Bekenstein-Hawking area law. Each measurement event contributes a discrete entropy increment proportional to the area imprinted on the horizon by the associated energy-momentum transfer.

This derivation parallels Jacob Bekenstein's original argument [9], in which the entropy increase of a black hole is inferred from the minimal energy required to localize and drop a system across the horizon; here, the same structure emerges from the minimal energy cost of registering information at the horizon scale.

The two bounds are saturated simultaneously when $a = c^2/R$ — which is exactly the condition for being *at a horizon.* The horizon is therefore not just where thermodynamics happens to work out, it is the unique locus where the minimum cost of information acquisition equals the maximum information content of the region.

## 2.6 Summary

Beginning from the single relation (3), and using only Newton's laws and the equivalence principle, we have recovered the Schwarzschild radius, the surface gravity, and the Bekenstein-Hawking area law as consequences of the observation $\rightarrow$ acceleration identification. These are standard results, but their derivation here from the physical accounting of information acquisition rather than from thermodynamic postulates is the point.

The thrust quantum introduced in this section, the physical carrier of the measurement-induced impulse, will be given a explicit microstate realization in the sections that follow. It is identified with one of a pair of entangled photons modeled as Aichelburg-Sexl pp-waves crossing past Rindler horizons in the Unruh picture. In this case the back reaction of a single force carrier in one wedge is correlated with an equal and opposite reaction in the other from a common source.

# 3 Bilocal Microstate Construction

From now on we use natural units $\hbar = c = 1$ unless otherwise specified; factors of $\hbar$ and $c$ are restored where they aid physical interpretation.

The arguments of Section 2 establish a heuristic relationship between information acquisition, acceleration, and gravitational effects, but leave open the question of what a single microstate looks like concretely. In standard treatments, the Unruh effect is derived by coupling a detector to a pre-existing acceleration and reading off a thermal response from the vacuum. Following [6], we invert this logic: we place an interaction in the past, where a source injects non-thermal entangled excitations into the field, and show that the observer's accelerated response can be driven by these source-induced excitations. The thermal ensemble then emerges as a coarse-grained aggregate of such microstates, rather than being a primitive property of the vacuum.

Each microstate, equation (23) upcoming, is individually non-thermal in the sense that it carries a definite Rindler frequency $\omega$, the frequency natural to $\mathcal{R}$ and $\mathcal{L}$, which differs from the Minkowski frequency seen by $\mathcal{P}$ due to the position-dependent redshift of the Rindler geometry. Thermality is not a property of any single microstate but of the ensemble; it emerges when one asks only which wedge a detection occurred in, discarding the specific microstate identity, and is a consequence of coarse-graining rather than a premise of the construction.

We now upgrade this construction from a spin-0 scalar field in the original source [6] to the spin-1 electromagnetic field with a superposition of helicity pairs in a Bell state, introducing one unknown binary degree of freedom, the helicity outcome, undetermined until measured, which is the single bit responsible for the remaining thermality in the ensemble. The photon carries one quantum of action $\hbar$ and energy $E = \hbar\omega$, identifying the thrust quantum of Section 2 with a single photon whose polarization outcome, upon detection, is the classical bit acquired by each observer. The Bell state is the natural completion of the CPT[2] symmetry between wedges, which by the Bisognano–Wichmann theorem [10] is implemented by the modular conjugation operator $J$ mapping the right wedge algebra to the left wedge algebra, and as we discuss in Section 5, the photons imprint a definite geometric signature onto the horizon.

## 3.1 A Source and Two Observers

Three massive bodies are placed at spacetime positions $(t, z) = (0, 1)$, $(0, -1)$, and $(-1, 0)$, named $\mathcal{R}$ - *Right*, $\mathcal{L}$ - *Left*, and $\mathcal{P}$ - *Past* respectively. $\mathcal{P}$ serves as the common causal ancestor of the accelerations of $\mathcal{R}$ and $\mathcal{L}$ (see Figure 2). $\mathcal{P}$ emits an entangled pair of spin-1 massless photons with momenta $\pm k$ and Rindler frequency $\omega = \omega_k$. The momenta are perfectly balanced, so the source exhibits no net back-reaction in this minimal model. The two photons propagate into the left and right Rindler wedges, where $\mathcal{L}$ and $\mathcal{R}$ detect them respectively, each registering a helicity eigenvalue and acquiring one classical bit of information.

The three-party structure $\mathcal{P}$–$\mathcal{R}$–$\mathcal{L}$ is a tripartite entanglement: neither $\mathcal{R}$ nor $\mathcal{L}$ alone holds sufficient information to reconstruct the emission event at $\mathcal{P}$, but their two complementary bits together do, making $\mathcal{P}$ the logical AND of the pair and providing a natural microstate-level realization of interior reconstruction.

## 3.2 Double-Null Gauge

The emission event naturally defines two null directions: the right-moving photon has wave-vector $k_R^\mu = (\omega, 0, 0, k)$ and the left-moving photon has wave-vector $k_L^\mu = (\omega, 0, 0, -k)$,

[2] PCT in Bisognano–Wichmann.

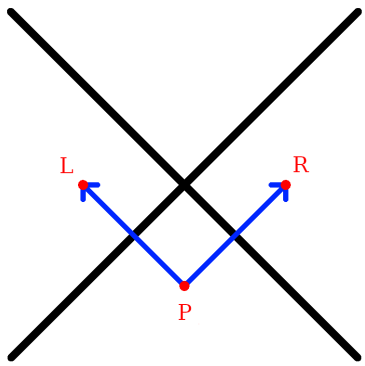


Figure 2: A source in the past, $\mathcal{P}$, emits an entangled photon pair that accelerates $\mathcal{R}$ into the right Rindler wedge and $\mathcal{L}$ into the left Rindler wedge. The two null directions of propagation define the $z^+$ and $z^-$ axes of the double-null coordinate system.

with equal and opposite spatial momenta. These two null directions are precisely the $z^+$ and $z^-$ axes of double-null coordinates

$$z^{\pm} = \frac{1}{\sqrt{2}}(t \pm z), \tag{13}$$

with transverse coordinates $\mathbf{z}_\perp = (x, y)$. Working in this frame we impose the symmetric double-null gauge condition

$$A_+(z) = 0, \qquad A_-(z) = 0, \tag{14}$$

so that the gauge field is supported entirely on the transverse components $\mathbf{A}_\perp = (A^x, A^y)$. This gauge respects the CPT symmetry that swaps the two wedges ($z^+ \leftrightarrow z^-$) simultaneously with ($R \leftrightarrow L$), and therefore treats both observers symmetrically by construction. The residual gauge freedom is fixed by requiring $\partial_i A^i = 0$ in the transverse plane, leaving precisely the two physical helicity degrees of freedom $\epsilon^i_{(\lambda)}$ with $\lambda = \pm 1$; consistency with Maxwell's equations and the absence of over-determination are verified in Appendix 7.

The lack of manifest four-dimensional covariance in equation (14) is a consequence of aligning the gauge with the pair of physical null directions selected by the source, rather than an arbitrary choice. The full Lorentz group is broken to the little group preserving the pair $\{k^\mu_R, k^\mu_L\}$, which is the two-dimensional Euclidean group $ISO(2)$ acting on the transverse plane. This is precisely the little group of massless particles, and the two helicity eigenstates $\lambda = \pm 1$ are its irreducible representations. All physical results are independent of the gauge choice.

## 3.3 The Bilocal Source

In double-null gauge the electromagnetic field is characterised by its transverse components alone. The bilocal driving source is symmetrized over both helicity assignments,

$$J^{ij}(x, y) = \lambda \left( \epsilon^{i*}_{(-1)} \epsilon^j_{(+1)} + \epsilon^{i*}_{(+1)} \epsilon^j_{(-1)} \right) f_L(x)\, f_R(y), \tag{15}$$

where $\lambda$ is a coupling constant, $f_L$ and $f_R$ are normalised mode profiles supported on the left and right wedges respectively, and $\epsilon^i_{(\lambda)}$ are the circular polarization vectors

$$\epsilon^i_{(+1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad \epsilon^i_{(-1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}. \tag{16}$$

The two terms in (15) are related by the CPT transformation in the sense of Bisognano–Wichmann [10], which maps the right wedge algebra to the left wedge algebra and simultaneously flips $\lambda \to -\lambda$. The symmetrization is therefore not an ad hoc choice but the unique assignment consistent with CPT symmetry and angular momentum conservation at the emission vertex; the total angular momentum vanishes in both terms.

The sourced Lagrangian reads

$$\begin{aligned}\mathcal{L}_{\text{sourced}} &= \mathcal{L}_{\text{free}} + \tfrac{1}{2}\lambda \int J^{ij}(x,y)\, A_i^+(x)\, A_j^+(y)\, d^4x\, d^4y \;+\; \text{h.c.} \\ &= \mathcal{L}_{\text{free}} + \tfrac{1}{2}\lambda \Big( a^{L\dagger}_{-k,(-1)}\, a^{R\dagger}_{k,(+1)} \;+\; a^{L\dagger}_{-k,(+1)}\, a^{R\dagger}_{k,(-1)} \;+\; \text{h.c.}\Big),\end{aligned} \tag{17}$$

where the two creation operator pairs correspond to the two CPT-related helicity assignments. The interaction term in (17) is a two-mode squeezing interaction at frequency $\omega$, now resolved by helicity and symmetrized over both assignments: it is the direct vector-field analogue of the scalar squeezing interaction [11], with the CPT symmetry between wedges made explicit. The photon pair carries total angular momentum zero in each term, consistent with emission from a scalar source at $\mathcal{P}$.

Each photon carries energy $E = \hbar\omega$ and one quantum of action $\hbar$, identifying the thrust quantum of Section 2 with a concrete physical object: the minimum actionable event is precisely a single photon. Its polarization, undetermined until measured, is the quantum bit whose outcome constitutes the classical record acquired by each observer upon detection.

The Unruh–Landauer bound is saturated not at the level of an individual photon but at the level of the source rate. The acceleration $a$ does not fix the frequency $\omega$ of each emitted quantum; rather, it determines the rate $\dot{N}$ at which the bilocal source must produce entangled pairs in order to fully[3] implement that acceleration, with the total power $\dot{N}\hbar\omega$ matching the Unruh thermal flux. Each individual photon carries exactly one quantum of action $\hbar$ and one binary helicity outcome — one bit in the sense of one binary alternative, carrying $\ln 2$ nats — and these are paired by the construction independently of $a$. The Landauer bound then operates as a consistency condition on the ensemble: the minimum energy cost per bit acquired, summed over the rate of acquisition, equals the power required to sustain the acceleration.

## 3.4 Polarization Bogoliubov Transformation

We now show that the polarization labels survive the Rindler–Unruh mode decomposition, so that the thermal ensemble retains its helicity structure and the anticorrelation of the emitted pair descends unambiguously to the level of individual Rindler microstates.

In the transverse plane, the electromagnetic field in double-null gauge decomposes into two independent scalar-like sectors labelled by helicity $\lambda = \pm 1$. For each sector, the right-wedge field expands as

$$A^i_\lambda(z) = \int_0^\infty \frac{d\omega}{2\pi}\, \frac{1}{\sqrt{2\omega}}\, \epsilon^i_{(\lambda)} \left[ b^R_{\omega,\lambda}\, u^R_\omega(z) + b^{R\dagger}_{\omega,\lambda}\, u^{R*}_\omega(z) \right], \tag{18}$$

where $u^R_\omega(z)$ are the right-wedge Rindler modes at Rindler frequency $\omega$, and $b^R_{\omega,\lambda}$, $b^{R\dagger}_{\omega,\lambda}$ are the corresponding annihilation and creation operators satisfying

$$\left[ b^R_{\omega,\lambda},\, b^{R\dagger}_{\omega',\lambda'} \right] = \delta(\omega - \omega')\, \delta_{\lambda\lambda'}. \tag{19}$$

An identical expansion holds in the left wedge with $R \to L$ and $\lambda \to -\lambda$, reflecting the anticorrelated helicity assignment. Because each helicity sector is independently a scalar-like sector, the Bogoliubov transformation relating Minkowski and Rindler modes takes

[3] The bilocal source may be only part of the acceleration, so $\dot{N}$ is actually a lower bound.

the same form in each sector. For helicity $\lambda$ at Rindler frequency $\omega$, the Unruh modes are

$$c_{\omega,\lambda} = \cosh\theta_\omega \; b^R_{\omega,\lambda} - \sinh\theta_\omega \; b^{L\dagger}_{\omega,-\lambda}, \tag{20}$$

where $\tanh\theta_\omega = e^{-\pi\omega c/a}$ encodes the Unruh distribution at acceleration $a$, and the CPT symmetry between wedges is reflected in the superposition over both helicity assignments in the Bell state. The transformation (20) is the standard Bogoliubov relation for a massless field [8, 12], now carrying a helicity label throughout. The Minkowski vacuum thermofield double state, in the Rindler basis reads, for each helicity sector,

$$|0\rangle^{(\lambda)}_M = \frac{1}{\cosh\theta_\omega} \sum_{n=0}^{\infty} \tanh^n\theta_\omega \; |n_{\omega,\lambda}\rangle_R \, |n_{\omega,-\lambda}\rangle_L, \tag{21}$$

and the full vacuum is the product over all $\omega$ and both helicities:

$$|0\rangle_M = \prod_{\omega,\lambda} |0\rangle^{(\lambda)}_M. \tag{22}$$

Equation (21) makes explicit that the Minkowski vacuum is a sum over entangled helicity-anticorrelated pairs at each Rindler frequency. The bilocal source selects a single term from this sum at fixed $\omega$ and fixed helicity pair $(\lambda, -\lambda) = (+1, -1)$, yielding the single-microstate

$$|\Psi_\omega\rangle_M = \frac{1}{\sqrt{2}} \left( |1_{\omega,+1}\rangle_R \, |1_{\omega,-1}\rangle_L + |1_{\omega,-1}\rangle_R \, |1_{\omega,+1}\rangle_L \right) + \mathcal{O}(\text{higher order}). \tag{23}$$

$\mathcal{R}$'s and $\mathcal{L}$'s helicity outcomes are undetermined until measurement, at which point they are perfectly anticorrelated by the Bell state structure: each observer acquires one classical bit, and the two bits are complements of one another.

The key structural result is that the helicity label $\lambda$ is conserved under the Bogoliubov transformation: it enters equation (20) as a spectator index and exits unchanged on the Unruh operators $c_{\omega,\lambda}$. Thermality, when it arises from coarse-graining over the ensemble of microstates (23), is therefore helicity-resolved: each helicity sector thermalises independently, and the anticorrelation between wedges is preserved at every level of the construction.

### 3.5 Summary

The passage from the scalar to the vector field construction introduces three structural improvements. First, the photon carrying $E = \hbar\omega$ and one quantum of action per cycle identifies the thrust quantum of Section 2 with a concrete physical object, closing the Landauer acquisition gap. Second, the double-null gauge condition, motivated by the pair of null directions defined by the emission event, treats both Rindler wedges symmetrically and exposes the CPT swap symmetry as the organising principle of the construction.

Third, the Bell state structure, conserved through the Bogoliubov transformation, leaves the helicity outcome undetermined until measurement, at which point each observer acquires one classical bit whose values are perfectly complementary by CPT symmetry: the two bits together constitute the correlated classical record that, in the AS pp-wave picture of Section 5, leaves a geometric trace on the horizon as information passes through it.

## 4 Matter as Substrate

The Unruh–Landauer relation of Section 2 constrains the minimal energy required to register information, but does not specify where that information resides physically. The

present section addresses this gap by identifying matter itself as the substrate in which such information is stored, with each act of acquisition corresponding to the deposition of a discrete quantum of action into the detector's degrees of freedom. We show how information is encoded in matter and energy and is reflected in the horizon geometry. This requires a physical account of what it means for matter to *carry* information.

A massive particle accumulates phase along its worldline at a rate set by its rest energy. The Compton frequency relation $f = E/h$ assigns a natural frequency to any mass $m$, and Bremermann's computational limit [13] independently recovers the same rate $f = mc^2/h$ as the maximum clock rate, or Nyquist rate, of any physical system with that energy $E$. These two results agree because they are both expressions of the same underlying fact: a quantum of action $\hbar$ is the minimal unit of physical evolution, and $mc^2/h$ is the rate at which a mass $m$ consumes such quanta.

Each such quantum corresponds to one sample[4] of the phase $e^{iS/\hbar}$ in the path integral, and classical trajectories emerge from the constructive interference of these samples across paths. Paths separated by less than one sample of action, less than $\hbar$, do not produce resolvable interference fringes: the phase difference is insufficient to complete a full cycle of constructive and destructive interference, so the trajectory cannot be distinguished at sub-sample resolution. The trajectory is resolved only down to the scale set by $\hbar/E$, which is precisely the Heisenberg uncertainty at energy $E$. This corresponds to a natural ultraviolet cutoff given by the energy $E$ of the system itself. Higher energies only come into play when external systems interact with the energy $E$ system.

This gives matter a precise sense in which it is a *substrate for information*: it accumulates samples, and each sample is a discrete unit of action, Lorentz-invariant and recorded in the particle's proper time. Matter is persistent memory in the operational sense that its history of samples is physically encoded in its state and cannot be erased without an energy cost governed by Landauer's principle. The maximum rate of this memory is bounded by Bremermann.

> *The signal being sampled at the Bremermann rate is the internal quantum state of the particle, its spin, helicity, and other quantum numbers, and the dynamic range is* $\log_2$ *of the dimension of the internal Hilbert space.*

It is worth pointing out that these concepts are additive. The union of two systems with masses $m_1$ and $m_2$ and internal Hilbert spaces of dimensions $d_1$ and $d_2$ has a sampling rate of

$$\frac{mc^2}{\hbar} = \frac{(m_1 + m_2)c^2}{\hbar} = \frac{m_1 c^2}{\hbar} + \frac{m_2 c^2}{\hbar}, \tag{24}$$

and a dynamic range of

$$\log_2(d) = \log_2(\dim(H_1 \otimes H_2)) = \log_2(d_1 d_2) = \log_2(d_1) + \log_2(d_2), \tag{25}$$

for joint mass $m = m_1 + m_2$ and joint dimension $d = d_1 d_2$. Both the sampling rate and the information capacity are therefore extensive quantities, consistent with their interpretation as properties of matter that compose naturally under union.

The distinction between massless and massive degrees of freedom is therefore operational: massless quanta transmit information, while massive systems retain it, accumulating the action quanta that constitute physical memory. A massless particle, the photon, has no rest frame, no proper time, and accumulates no action between emission and absorption. It does not sample. It therefore carries no persistent memory and cannot serve as a substrate. Its role in the present construction is precisely that of a pure information carrier. A single photon carries exactly two internal irreducible constituents, distinct in kind:

[4] We use *sample* throughout to denote a single zero crossing of the phase $e^{iS/\hbar}$; readers may equivalently read this as *event* or *tick*. It is only defined upto the Heisenberg resolution.

- **One quantum of action** $\hbar$: the physical transfer, deposited as energy $\hbar\omega$ into the detector, incrementing its Bremermann sample rate by $\Delta f = \hbar\omega/h$ for as long as the energy remains in the mass. This increment is the physical trace of the interaction in the detector's matter-memory.
- **One bit of polarization**: the informational outcome, the classical record of which helicity, $\lambda = +1$ or $\lambda = -1$ — was absorbed. This bit persists in the detector's state for as long as the detector retains the record. Erasure of this bit carries a minimum energy cost governed by Landauer's principle, regardless of how the record was originally established.

## 4.1 Types of sample

From a given observer's perspective, some samples appear to carry information while others do not. One can describe three types of samples in terms of how the observer registers or retains their outcomes.

- **Absorption samples** are measurement events: a photon is detected, one quantum of action $\hbar$ is deposited persistently into the detector's matter-memory, and one helicity outcome is written as a classical bit. The Landauer cost $\Delta E \geq k_B T \ln 2$ is paid immediately and in full on acquisition. This is the sample category that drives the bilocal construction: $\mathcal{R}$ and $\mathcal{L}$ each undergo exactly one absorption sample per emitted pair, and it is these samples whose gravitational imprint is modelled by the AS pp-wave in Section 5.
- **Free samples** are pure phase accumulation events: a massive particle propagates along its worldline, incrementing its sample count at the Bremermann rate $mc^2/h$, without writing any bit into any detector's state. No Landauer cost is incurred, no classical record is produced, and no observer acquires information. These are the samples that constitute mass and proper time in the background sense, the substrate on which absorption and emission events act.
- **Emission samples** are the time-reverse of absorption: a bit stored in matter-memory is released, the corresponding quantum of action is re-emitted as radiation, and the Landauer cost is paid on the way *out* rather than on the way in. This is the category Landauer's principle most directly governs in its original erasure formulation.

A moving mirror realizes absorption and emission as reflection: it receives a photon, stores the helicity bit transiently in its matter degrees of freedom during a period of free sample accumulation, and re-emits the photon, paying the erasure cost at emission. An ensemble of such mirrors, say two silvered mirrors back to back, can cycle through absorption, free accumulation, and emission without ever producing a permanent classical record.

It is worth noting that the absorption/free/emission classification is observer-relative in the same sense as quantum state reduction itself. What constitutes an irreversible absorption sample for one observer may be a free sample for an external observer. If $\mathcal{R}$ truly forgets the helicity outcome, in the sense that his matter-memory becomes disentangled from the bit, the photon can interfere with its partner at a future $\mathcal{P}$. The classification is not a property of the interaction itself but of the observer's epistemic access to the outcome. True irreversibility is approached asymptotically as the bit becomes correlated with an increasing number of environmental degrees of freedom, which is the standard decoherence picture now applied to the sample classification.

The $\hbar$ is what makes the sample physical; the bit is what makes it meaningful to an observer. These are distinct: the action quantum is deposited regardless of whether a persistent record is produced, while the bit exists only relative to an observer for whom

the outcome is accessible. That a single photon carries one of each in the minimal case is the reason it is the natural unit of the construction.

## 4.2 Penrose's Zig-Zag Mechanism

The matter substrate argument of the preceding subsection established that massive particles sample at the Bremermann rate $mc^2/h$ and that each sample is a quantum of action $\hbar$ accumulated in the path integral phase $e^{iS/\hbar}$. But it left open a prior question: *why* does matter sample at all? The zig-zag[5] construction of Penrose and Dirac supplies a concrete physical answer, and in doing so reveals that the bilocal microstate construction of Section 3 is structurally identical to it at the level of individual events.

### 4.2.1 The Dirac 4-Spinor as a Bilocal System

The Lorentz algebra decomposes as

$$\mathfrak{so}(3,1) \cong \mathfrak{su}(2) \times \mathfrak{su}(2), \tag{26}$$

where the two $\mathfrak{su}(2)$ factors correspond to the two helicities: the first is generated by boosts combined with counter-clockwise rotations, and the second by boosts combined with clockwise rotations [14]. A Dirac 4-spinor decomposes under this algebra into two Weyl 2-spinors of opposite helicity $\lambda = \pm\frac{1}{2}$ and equal and opposite momenta. Penrose [15] proposes a physical interpretation of this decomposition: the two 2-spinors are real physical objects, each moving instantaneously at the speed of light in opposite directions, interacting at regular intervals via the Higgs field at the De Broglie frequency.

The structural parallel with the bilocal construction is very close:

- The two Weyl 2-spinors, moving at $c$ in opposite directions with opposite helicity, correspond to the two photons emitted by $\mathcal{P}$ into the left and right Rindler wedges.
- The Higgs field plays the role of $\mathcal{R}$ and $\mathcal{L}$: it is the common mediator whose interaction reflects the entangled pair and sets the interaction rate.
- Between Higgs interactions, each 2-spinor is massless and moves at $c$, carrying no persistent memory, pure information carriers, exactly as the photons in the bilocal construction.
- The Higgs interaction event itself is the measurement event: it deposits one quantum of action $\hbar$, flips the active helicity from $+\frac{1}{2}$ to $-\frac{1}{2}$ and back, and produces one sample of proper time.

Mass emerges from the regularity of these interactions: a particle is massive because it undergoes repeated Higgs interactions at the De Broglie frequency, and its mass is proportional to that frequency via $E = \hbar\omega$. The Bremermann sample rate $mc^2/h$ is therefore not a coincidence or a heuristic, it is the Higgs interaction rate, the rate at which the zig-zag mechanism deposits quanta of action into the particle's proper time.

### 4.2.2 Mass as Accumulated Measurement History

Between Higgs interactions, the two 2-spinors are massless and moving at $c$ in opposite null directions. Each spinor therefore has a local Rindler structure relative to the other: from the perspective of the right-moving spinor, the left-moving spinor is behind a causal horizon, and vice versa. The Higgs interaction is the event that collapses this local horizon and produces one sample, one unit of proper time, exactly as the photon absorption by $\mathcal{R}$ collapses the Rindler horizon and produces one sample in the bilocal construction.

[5] Penrose calls these particle *zig-zag* particles, we are tempted to call them *tick-tock* particles, but only in a footnote.

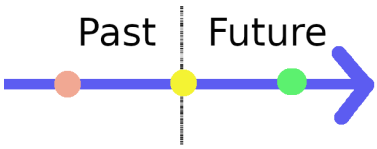


Figure 3: A 1-dimensional (proper) time line. The red, yellow, and green dots are 0-dimensional events, in the past, present and future respectively. A screen is pictured at the present.

This gives a precise sense in which mass is an accumulated measurement history. Each Higgs interaction is a measurement event: it deposits one quantum of action $\hbar$ into the particle's matter-memory, persistently incrementing its sample rate by $\Delta f = \Delta E/h$. The particle's mass is the total accumulated record of all such interactions since its formation. Matter is not just the substrate for information; its mass encodes the accumulated history of its Higgs interaction events.

### 4.2.3 Proper Time as a Foliation of Micro-Horizons

The series of Higgs interaction events along a particle's worldline is a one-dimensional foliation of the proper time line by discrete events, each of which is a micro-horizon separating the causal past from the causal future of that interaction, see Figure 3. This is the one-dimensional analogue of the Rindler horizon separating left and right wedges in the bilocal construction, and it matches the holographic screen picture of Verlinde [2]: each screen is a surface at which information is registered and a sample of proper time is produced.

The zig-zag mechanism therefore provides a physical grounding for the matter substrate argument at the level of individual particles. The bilocal construction of Section 3 is the macroscopic version of the same structure: $\mathcal{R}$ and $\mathcal{L}$ play the role of the Higgs field, the two photons play the role of the two Weyl spinors, and they produce samples of proper time in the zig-zag picture. The difference in scale is large but the logical structure is identical.

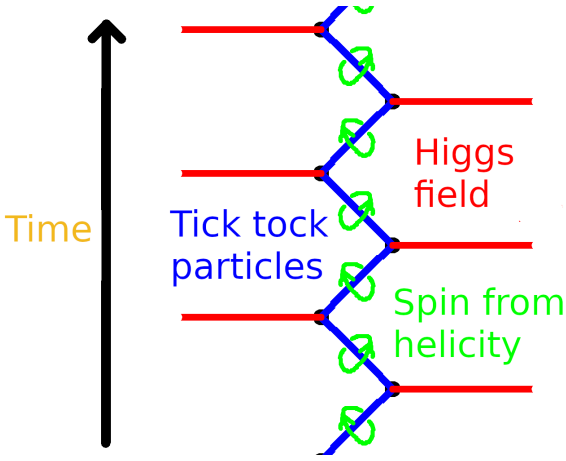


Figure 4: These figures were adapted from [15]. They show his interpretation of the Dirac equation and the Higgs mechanism for an electron.

## 4.3 Matter as a Physical Channel

The sampling rate and dynamic range of matter can be combined into a channel capacity in the Shannon sense. For a system of mass $m$ with internal Hilbert space dimension $d$, the maximum rate of information flow is

$$C = \underbrace{\frac{mc^2}{h}}_{\text{bandwidth}} \times \underbrace{\log_2(d)}_{\text{dynamic range}} \qquad \text{(bits per second)}, \tag{27}$$

where the bandwidth is the Bremermann sampling rate and the dynamic range is the information content per sample. The product $C$ is the *channel capacity* of the matter substrate: the maximum rate at which physical interactions can write information into its degrees of freedom.

The three sample types of Section 4 correspond to three distinct regimes of the data processing inequality $H(\text{out}) \leq H(\text{in})$, which states that information cannot be created by physical processes, only preserved or lost:

$$\text{Free:} \quad H(\text{out}) = H(\text{in}), \tag{28}$$

$$\text{Absorption:} \quad H(\text{field, out}) < H(\text{field, in}), \tag{29}$$

$$\text{Emission:} \quad H(\text{detector, out}) < H(\text{detector, in}). \tag{30}$$

Free samples saturate the inequality: no information is lost or gained, the channel runs at capacity without transmitting anything. Absorption samples strictly decrease the field's information content as it flows into the detector: the channel is being *used*, information is being written into matter-memory at a rate bounded by $C$. Emission samples strictly decrease the detector's information content as the stored bit is released back into the field: the channel runs in reverse, and the Landauer cost is the energy price of that reversal.

The data processing inequality is therefore the information-theoretic statement of the thermodynamic arrow of time in this framework: absorption and emission are the two directions of persistent information flow, and free samples are the reversible background against which both are measured. The full quantum channel structure of the Bogoliubov transformation, its classical capacity in the Holevo–Schumacher–Westmoreland, we leave for future work.

The three fundamental constants $c$, $\hbar$, and $G$ play precisely the roles of propagation speed, sampling rate, and dynamic range in the channel capacity picture. The speed of light $c$ sets the causal propagation speed of signals, identifying space with time at the rate m/s. The reduced Planck constant $\hbar$ sets the sampling rate via the Compton frequency $mc^2/\hbar$, the temporal bandwidth of the matter channel. The gravitational constant $G$, through the Schwarzschild radius $r_s = 2Gm/c^2$, sets the dynamic range: the maximum energy storable in a spatial region without gravitational collapse, bounding the information capacity per sample. Their combination $\ell_P^2 = \hbar G/c^3$ is the Planck area, the minimal spatial pixel per independent action tick, and counting such pixels on a surface reproduces the Bekenstein bound. The three constants are therefore not independent inputs to the channel capacity but its three defining parameters: propagation, resolution, and dynamic range.

The Bekenstein and Bremermann bounds are therefore the same constraint viewed from orthogonal directions: Bekenstein counts information capacity spatially, as area in Planck units, while Bremermann counts it temporally, as samples per second. Integrating the channel capacity over the natural causal timescale $\tau \sim Gm/c^3$ gives

$$C \cdot \tau = \frac{mc^2}{h} \cdot \frac{Gm}{c^3} = \frac{Gm^2}{hc} \sim H_{\text{Bekenstein}}, \tag{31}$$

up to numerical factors and the dynamic range $\log_2(d)$. The spacelike and timelike information bounds are therefore consistent faces of the same underlying constraint.

## 4.4 Horizon Separation

The horizon enters this picture not as a storage medium but as a boundary condition. It marks the surface at which the matter-memory on one side is causally disconnected from the matter-memory on the other. Horizon entropy, in this account, is not a property of the horizon surface itself but a count of the samples that have been written into the matter on each side, bounded by Bremermann from above and by Landauer from below. The holographic appearance of this count, proportional to area rather than volume, is a consequence of the Unruh–Landauer and Bekenstein relations derived in Section 2 for a spherically symmetric mass distribution and does not require holography as an independent postulate.

This reframing has a direct consequence for the emergent spatial separation developed in Section 6 below. The correlated measurement outcomes of $\mathcal{R}$ and $\mathcal{L}$ are not encoded in the horizon geometry in the first instance, they are encoded in the matter of their respective detectors, each of which has absorbed one photon and increased its Bremermann sampling rate by $\Delta f = \hbar\omega/h$. The geometric correlations between the two wedges, realized as correlated AS pp-wave imprints in Section 5, are downstream of this matter-level encoding. The bridge, if it exists, is built from correlated samples, not from the horizon itself.

# 5 pp-Wave Geometry

WLOG we introduce coordinates for the photon traveling from $\mathcal{P}$ $(-1, 0)$ to $\mathcal{R}$ $(0, 1)$. Let $u = t - z + 1$ and $v = t + z + 1$, where the photon propagates along the null plane $u = 0$, $z$ is the longitudinal direction, and $(x, y)$ are transverse coordinates. Then $(u, v) = (0, 0)$ is $\mathcal{P}$'s location, $(u, v) = (0, 2)$ is $\mathcal{R}$'s location, and $(u, v) = (0, 1)$ is the intersection of the photon with the past right Rinder horizon. The situtation is reversed for the left moving photon, we swap $u$ and $v$ to describe photon moving from $\mathcal{P}$ to $\mathcal{L}$.

## 5.1 The AS pp-Wave and Its Effects

Test particles at fixed transverse positions $(x, y)$ experience two effects as the shock front crosses $u = 0$:

1. **Longitudinal shift.** The $v$ coordinate is displaced by

$$\Delta v \propto -\ln r^2, \qquad r^2 = x^2 + y^2. \tag{32}$$

   This shift is lightlike: no proper time elapses for a particle crossing the shock, but it is displaced along the null direction. Near the axis ($r \to 0$) the shift diverges, encoding the concentration of energy along the photon's trajectory.

2. **Transverse focusing.** The gradient of the longitudinal shift produces a transverse velocity

$$\Delta \mathbf{v}_\perp \propto \nabla \ln r \propto \frac{1}{r}, \tag{33}$$

   directed toward the axis. Test particles are pulled toward the centre of the wave, regardless of their initial transverse separation.

## 5.2 The Longitudinal Shift Makes Space

The longitudinal shift deserves closer attention, because it carries a meaning beyond a mere coordinate displacement. The photon emitted by $\mathcal{P}$ propagates along the null direction; as it crosses the Rindler horizon, the AS shift displaces the future light cone of every test particle it passes. For the photon traveling into the right wedge, the shift

pushes the right wedge's geometry in the $+v$ direction; for the entangled partner traveling into the left wedge, the shift pushes the left wedge in the $-v$ direction. The two wedges are displaced *away from each other* along the longitudinal null direction. See Figure 5.

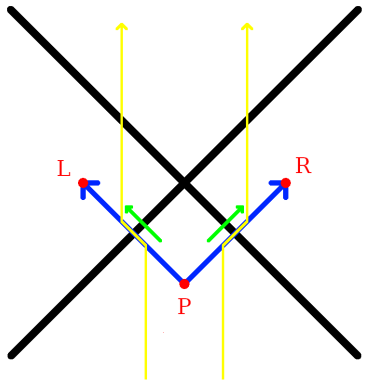


Figure 5: The green arrows show the longitudinal shift that occurs where the photons cross the horizon. Test particles, shown in yellow, are pushed outwards away from $z = 0$ due to the creation of space along the null lines originating from $\mathcal{P}$.

At the intersection of the shock front $u = 0$ with the past right Rindler horizon at $(u, v) = (0, 1)$, the transverse $(x, y)$ plane is identified with the horizon area elements $dA = dx\,dy$. The longitudinal shift $\Delta v \propto -\ln r^2$ displaces the horizon outward in the $v$ direction, pushing the right wedge geometry away from the horizon. The entangled partner photon produces a symmetric shift in the $u$ direction at the left Rindler horizon, pushing the left wedge away from its horizon. The two wedges are therefore displaced away from their respective horizons in opposite null directions.

## 5.3 Transverse Focusing Increments Horizon Area

The transverse focusing of the AS pp-wave produces a calculable increment in horizon area. We derive this using the Raychaudhuri equation on the null congruence generating the Rindler horizon, following Jacobson [1], but applied here to a single microstate rather than a thermodynamic average.

For a null congruence with generators $k^\mu$ and expansion $\theta$, the Raychaudhuri equation gives

$$\frac{d\theta}{du} = -\tfrac{1}{2}\theta^2 - \sigma_{\mu\nu}\sigma^{\mu\nu} - 8\pi G\, T_{\mu\nu}k^\mu k^\nu. \tag{34}$$

Integrating over the horizon patch in the linearised regime and using $k^\mu = \delta^\mu_u$, the area change is

$$\Delta A = 8\pi G \int T_{uu}\, du\, d^2x_\perp. \tag{35}$$

In a strict AS construction, the source is a point particle and $T_{\mu\nu}$ is distributional, concentrated on the axis. The present construction provides a natural regularization: the photon is a normalizable mode of the electromagnetic field in double-null gauge, with a smooth transverse profile set by the wavelength $\sim c/\omega$. The electromagnetic stress-energy $T_{uu}$ is well-defined throughout, integrates to $\hbar\omega$ over the transverse plane by energy conservation, and recovers the strict AS limit as $\omega \to \infty$ with the logarithmic divergence regulated at the scale $c/\omega$.

Substituting into equation (35) gives

$$\boxed{\Delta A = 8\pi G\,\hbar\omega = 8\pi G\,\Delta E.} \tag{36}$$

The horizon area grows by one discrete, calculable increment per photon absorbed, microstate by microstate, without reference to bulk thermodynamic averages.

Applying the Bekenstein–Hawking relation as a heuristic, valid for Schwarzschild but taken here as an ansatz for the Rindler case, yields a corresponding entropy increment

$$\Delta S = \frac{\Delta A}{4\ell_P^2} = \frac{2\pi\,\Delta E}{\hbar\omega/\hbar} = 2\pi, \tag{37}$$

in natural units, consistent with one Planck-area patch per detection event.

The longitudinal shift $\Delta v \propto -\ln r^2$ produced by each AS pp-wave at the Rindler horizon is a supertranslation in the sense of Bondi–Metzner–Sachs (BMS) symmetry. Hawking, Perry, and Strominger [16] showed explicitly that a black hole is supertranslated by the passage of an asymmetric shockwave, and the AS longitudinal shift is precisely the linearised supertranslation field on the horizon. Carlip [17] demonstrated that the near-horizon diffeomorphisms of a generic black hole are enhanced to a $\mathrm{BMS}_3$ algebra whose central charge determines the Bekenstein–Hawking entropy, but his programme identifies the symmetry without specifying the physical process that excites individual modes. The bilocal construction supplies that process: each measurement event produces one AS pp-wave, which deposits one supertranslation at the Rindler horizon, exciting one mode of the $\mathrm{BMS}_3$ algebra. The entropy count is consistent with Carlip's result, the microstates being counted are the measurement events themselves.

## 5.4 Action and the Gravitational Boundary

Each absorption sample deposits one quantum of action $\hbar$ into the detector's matter-memory. The total action accumulated across all measurement events is therefore

$$S_{\text{matter}} = \sum_i \hbar\omega_i, \tag{38}$$

where the sum is over all absorption samples, each contributing energy $\hbar\omega_i$ to the detector mass. This is the matter action counted sample by sample: one quantum of action per event, one event per emitted pair.

The same events have a geometric description. By eq. (36), each absorption sample contributes a horizon area increment $\Delta A_i = 8\pi G\hbar\omega_i$, so the total area increment summed over all events is

$$\sum_i \Delta A_i = 8\pi G \sum_i \hbar\omega_i = 8\pi G\, S_{\text{matter}}. \tag{39}$$

This geometric sum has a precise identification in the gravitational action. On a stationary Rindler horizon the unperturbed expansion of the null generators vanishes, $\theta = 0$, so the Gibbons–Hawking–York boundary term [18] is sourced entirely by the AS pp-wave perturbations:

$$S_{GHY}\big|_{\mathcal{H}} = \frac{c^4}{8\pi G}\int_{\mathcal{H}} \theta\, d^2x_\perp\, du = \frac{c^4}{8\pi G}\sum_i \Delta A_i = \frac{c^4}{8\pi G}\cdot 8\pi G \sum_i \hbar\omega_i = \sum_i \hbar\omega_i = S_{\text{matter}}. \tag{40}$$

The matter action and the gravitational boundary action are therefore equal, event by event, on the stationary Rindler horizon. This is not a bulk result: the bulk Einstein–Hilbert action vanishes on-shell, and it is the boundary term that carries the physical content. The microstate construction identifies what that boundary content is: each contribution to $S_{GHY}$ corresponds to one quantum of action deposited into matter-memory by one measurement event.

This result is consistent with and extends programmes of Jacobson [1] and Padmanabhan [19], which argue that the gravitational boundary term rather than the bulk Einstein–Hilbert action is the fundamental gravitational degree of freedom. Those programmes

establish this at the level of thermodynamic averages. The present construction realises the same identification at the level of individual microstates, with each boundary contribution sourced by a specific quantum measurement event rather than a bulk average. The standard result emerges from summing over the ensemble; the microstate picture supplies what the thermodynamic programmes leave open: the physical process underlying each boundary contribution.

# 6 Entanglement as Spatial Information

The bilocal source prepares an anticorrelated Bell state (23). We consider the simplest case: both $\mathcal{R}$ and $\mathcal{L}$ measure in the same basis and both obtain a definite outcome, which we label $H_+$ or $H_-$. In this case their outcomes are perfectly anticorrelated by the Bell state structure, independently of which basis is chosen. The question

*"who observed $H_+$?"*

then has a binary answer: $\mathcal{L}$ (left wedge) or $\mathcal{R}$ (right wedge). This is a *spatial bit*, a bit that encodes not a polarization state but a location in space.

Each measurement event has exactly two irreducible constituents: one quantum of action $\hbar$ written persistently into the detector's matter-memory as a series of samples, and one bit of polarization registered as a classical outcome. The entanglement between $\mathcal{R}$ and $\mathcal{L}$ is consumed in producing these two records, an increase in sampling rate and one bit at each wedge. What remains after measurement is not the entanglement itself but its classical residue: the perfect anticorrelation between the two bits, and the correlated pair of samples written into the matter on each side of the horizon. The spatial bit is precisely the polarization bit now read as a location rather than a spin state: the same bit that records *what* was observed also records *where*, because in this construction the two wedges carry opposite helicities by design. This double duty is forced: the anticorrelated helicity assignment of the bilocal source makes the two wedges distinguishable by construction, so the spatial bit is already latent in the source geometry before any measurement occurs.

The information is carried by the matter of the detectors, not by the horizon. As established in Section 4, the horizon is a boundary that makes the left/right distinction sharp; it is the matter on each side that accumulates the samples and holds the classical record. $\mathcal{R}$'s detector has absorbed one photon, increased its sampling rate, and registered polarization $H_+/H_-$. $\mathcal{L}$'s detector has absorbed one photon, increased its sampling rate, and registered polarization $H_-/H_+$. The two records are perfectly complementary by the anticorrelated structure of the bilocal source, and their complementarity is encoded in matter before it is reflected in geometry.

## 6.1 ER = EPR

The Unruh vacuum is the thermofield double state of the left and right Rindler wedges [20]: equation (21) is precisely this state, with the helicity label $\lambda$ carried through. This places the present construction squarely within the ER=EPR setting of Maldacena and Susskind, the ER=EPR correspondence is the ambient mathematical structure within which the bilocal construction operates.

The bilocal source selects one term from the thermofield double sum at fixed $\omega$ and fixed helicity pair, yielding the single microstate (23). The EPR side is the entangled photon pair emitted by $\mathcal{P}$, whose quantum correlations are consumed by the two measurement events. The ER side is the pair of correlated AS longitudinal shifts in opposite null directions: the geometric imprint of one term in the thermofield double sum, calculable and explicit. The spatial bit, the answer to "who observed $H_+$?", is the classical record of which side of the bridge each measurement occurred on. The bridge is not a

stand-in for the entanglement but its physical instantiation, built from correlated matter samples, one quantum of action at a time.

What is new here is that both sides of the correspondence are calculable at the single-microstate level, with a definite energy-momentum source and a definite geometric output, rather than established by a bulk thermodynamic argument. The anticorrelated helicity structure gives the bridge an orientation absent in the scalar case: the two sides carry opposite handedness, and whether this constitutes a topological invariant of the transverse geometry is an open question we leave for future work.

This completes the microstate programme. Each entropy increment in the Bekenstein–Hawking area law corresponds to a single AS pp-wave interaction. Each pp-wave interaction corresponds to a single measurement event in the bilocal construction. Each measurement event both encodes a spatial bit and generates the spatial separation that bit refers to. The gravitational, thermodynamic, information-theoretic, and geometric descriptions of the horizon are unified at the level of individual microstates, and thermality, entropy, and space itself emerge together from the coarse-grained aggregate of these events.

# 7 Appendix: Consistency of the Double-Null Gauge Condition

We verify that the double-null gauge condition $A_+ = A_- = 0$ is consistent with Maxwell's equations and leaves exactly two physical degrees of freedom.

## 7.1 Coordinates and Notation

In double-null coordinates $(x^+, x^-, x^1, x^2)$ the Minkowski metric reads

$$ds^2 = -2\,dx^+\,dx^- + (dx^1)^2 + (dx^2)^2, \tag{41}$$

so that $\partial_+ = \partial/\partial x^+$, $\partial_- = \partial/\partial x^-$, and the d'Alembertian is

$$\Box = -2\partial_+\partial_- + \partial_i\partial^i, \qquad i = 1, 2. \tag{42}$$

The gauge field has components $A_\mu = (A_+, A_-, A_1, A_2)$ and transforms as

$$A_\mu \to A_\mu + \partial_\mu \Lambda \tag{43}$$

under a gauge transformation with parameter $\Lambda(x^+, x^-, x^1, x^2)$.

## 7.2 Imposing the Gauge Conditions in Stages

The single gauge parameter $\Lambda$ is a function of all four coordinates. We use it in three stages, each consuming a different functional dependence.

**Stage 1: $A_+ = 0$.** Requiring $A_+ + \partial_+\Lambda = 0$ determines $\Lambda$ as a function of $x^+$:

$$\Lambda(x^+, x^-, x^i) = -\int^{x^+} A_+(s, x^-, x^i)\,ds + \Lambda_0(x^-, x^i), \tag{44}$$

where $\Lambda_0(x^-, x^i)$ is an arbitrary residual function independent of $x^+$.

**Stage 2: $A_- = 0$.** Under the residual freedom $\Lambda_0(x^-, x^i)$, the $A_-$ component transforms as $A_- \to A_- + \partial_-\Lambda_0$. Setting this to zero determines $\Lambda_0$ as a function of $x^-$:

$$\Lambda_0(x^-, x^i) = -\int^{x^-} A_-(s, x^i)\,ds + \Lambda_1(x^i), \tag{45}$$

where $\Lambda_1(x^i)$ is a residual function of the transverse coordinates only.

**Stage 3: Transverse Coulomb condition $\partial^i A_i = 0$.** Under the remaining freedom $\Lambda_1(x^i)$, the transverse components transform as $A_i \to A_i + \partial_i \Lambda_1$. Imposing $\partial^i A_i = 0$ requires

$$\nabla_\perp^2 \Lambda_1 = -\partial^i A_i, \tag{46}$$

which is a Poisson equation in the transverse plane with a unique solution (up to harmonic functions, fixed by boundary conditions). This exhausts the gauge freedom entirely.

The three conditions have consumed the full gauge parameter $\Lambda$ in stages, with no over-determination: each stage uses a strictly different functional dependence of $\Lambda$.

## 7.3 Physical Degrees of Freedom

After imposing $A_+ = A_- = 0$ and $\partial^i A_i = 0$, the remaining components are $A_1$ and $A_2$ subject to one differential constraint $\partial^i A_i = 0$. In momentum space this reads $k^i A_i = 0$.

For our specific construction, the photons propagate in the $\pm z$ direction, so their wave-vectors are purely null: $k_\perp^i = 0$ for $i = 1, 2$. The transverse Coulomb condition $k^i A_i = 0$ is therefore trivially satisfied for any $A_1, A_2$, and both transverse components are independent. The two circular polarization vectors

$$\epsilon^i_{(+1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}, \qquad \epsilon^i_{(-1)} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} \tag{47}$$

are the two independent physical degrees of freedom, corresponding to helicity $\lambda = \pm 1$. This is the correct count for a massless spin-1 field in four dimensions.

The trivial satisfaction of the transverse Coulomb condition relies on the photons propagating purely in the $\pm z$ direction with vanishing transverse momentum $k_\perp^i = 0$. For photons with nonzero transverse momentum the condition $k^i A_i = 0$ imposes a non-trivial constraint and the degree of freedom count requires revisiting.

## 7.4 Consistency with Maxwell's Equations

With $A_+ = A_- = 0$ and $\partial^i A_i = 0$, the non-zero field strength components are

$$F_{+i} = \partial_+ A_i, \qquad F_{-i} = \partial_- A_i, \qquad F_{ij} = \partial_i A_j - \partial_j A_i, \qquad F_{+-} = 0. \tag{48}$$

Maxwell's equations $\partial^\mu F_{\mu\nu} = J_\nu$ decompose as follows.

**Transverse components $\nu = i$:**

$$-2\partial_+ \partial_- A_i + \partial^j (\partial_j A_i - \partial_i A_j) = J_i. \tag{49}$$

Using $\partial^i A_i = 0$, the second term simplifies to $\partial^j \partial_j A_i$, giving the standard wave equation

$$\boxed{\Box A_i = J_i.} \tag{50}$$

**Null components $\nu = \pm$:**

$$-\partial^i \partial_\pm A_i = J_\pm. \tag{51}$$

These are constraint equations relating the null source components to the transverse field. For our bilocal source, which is purely transverse, $J_+ = J_- = 0$, and the constraints reduce to

$$\partial^i \partial_\pm A_i = 0. \tag{52}$$

Differentiating the transverse Coulomb condition $\partial^i A_i = 0$ with respect to $x^\pm$ gives $\partial^i (\partial_\pm A_i) = 0$, so these constraints are automatically satisfied. ✓

### 7.5 Summary

The double-null gauge condition $A_+ = A_- = 0$ supplemented by the transverse Coulomb condition $\partial^i A_i = 0$:

1. uses the gauge freedom of $\Lambda$ completely and without over-determination, via a three-stage procedure consuming the $x^+$-, $x^-$-, and $x^i$-dependent parts of $\Lambda$ in sequence;
2. leaves exactly two physical degrees of freedom, the helicity eigenstates $\lambda = \pm 1$, consistent with the known count for a massless spin-1 field;
3. reduces Maxwell's equations to the standard transverse wave equation $\Box A_i = J_i$, with the null constraint equations automatically satisfied for a purely transverse source.

The gauge is therefore internally consistent and does not over-constrain the physical degrees of freedom. The CPT wedge-swap symmetry is preserved throughout: conditions on $A_+$ and $A_-$ are imposed symmetrically, and neither null direction is privileged over the other.

# 8 Conclusion

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